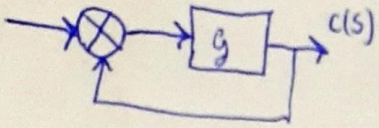
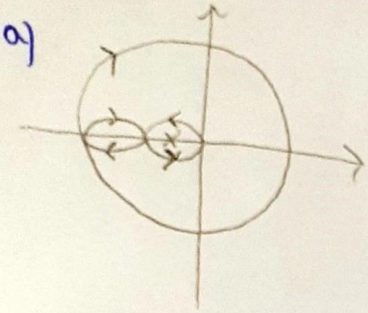
Nome: Werikson Frederico de O. Alves
Data: 21/09/2021
matrícula: 96708
P2 - Elt331 - Controle 2

①



Possui um polo no semiplano direito (P=1)

a)



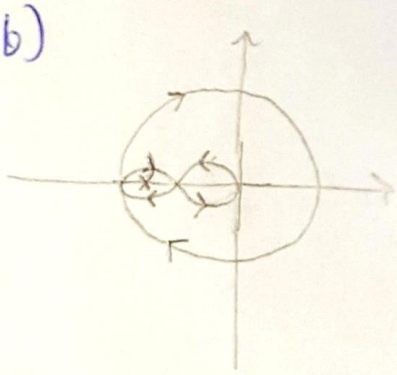
N = 1-1 = 0 (sentido horário e antihorário)

P = 1

instável, pois o gráfico possui 2 envoltórios, um no sentido horário e um no sentido anti-horário, e possui um polo no semiplano direito, logo, Z é diferente de zero.

$Z = N + P = (1-1) + 1 = 1$ //

b)



N = 2 (2 envoltórias no sentido horário)

P = 1

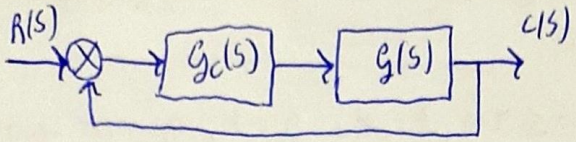
instável, pois pelo gráfico, percebe-se a presença de 2 envoltórios no sentido horário e como existe um polo no semiplano direito, o número de zeros (Z) é diferente de zero.

$Z = N + P = 2 + 1 = 3$ //

Weirison Alves

②

$$G(s) = \frac{5}{s(s+1)}$$



a) Avanço de fase, $MF \geq 50°$, $e_{ss} = 2\%$ (rampa unitária), usar 5° no ângulo.

$$\text{i)} \begin{cases} G_c(s) = K_c\alpha\dfrac{(Ts+1)}{(T\alpha s+1)} = K\dfrac{(Ts+1)}{(T\alpha s+1)} \; ; \; K = K_c\alpha \\ G_c(s)G(s) = K\dfrac{(Ts+1)}{(T\alpha s+1)} \cdot \dfrac{5}{s(s+1)} = \dfrac{(Ts+1)}{(T\alpha s+1)} \cdot G_1(s) \; ; \; G_1(s) = \dfrac{5K}{s(s+1)} \end{cases}$$

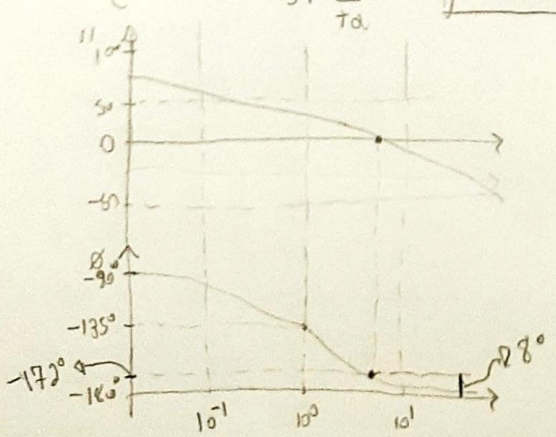
$$\text{ii)} \begin{cases} e_{ss} = 0{,}02 \Rightarrow K_v = \dfrac{1}{e_{ss}} \Rightarrow \boxed{K_v = 50} \\ K_v = \lim\limits_{s\to 0}\left[s\dfrac{(Ts+1)}{(T\alpha s+1)} \cdot \dfrac{5K}{s(s+1)}\right] \Rightarrow 50 = \dfrac{1}{1} \cdot \dfrac{5K}{1} \Rightarrow \boxed{K = 10} \Rightarrow \end{cases}$$

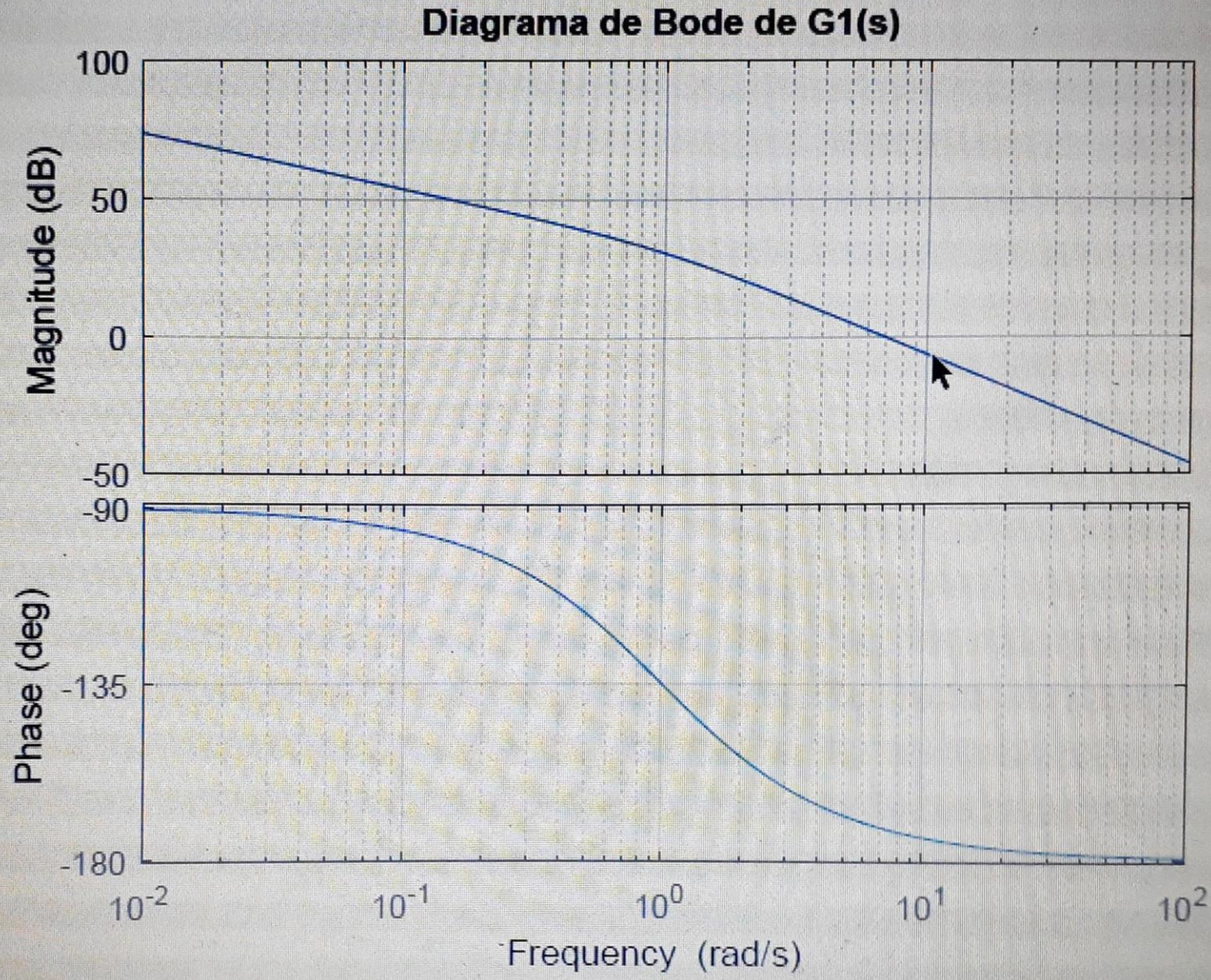
$$\text{iii)} \begin{cases} G_1(s) = \dfrac{50}{s(s+1)} \Rightarrow G_1(j\omega) = \dfrac{50}{\omega\sqrt{\omega^2+1}} \angle 0° - 90° - \operatorname{tg}^{-1}\left(\dfrac{\omega}{1}\right) \end{cases}$$

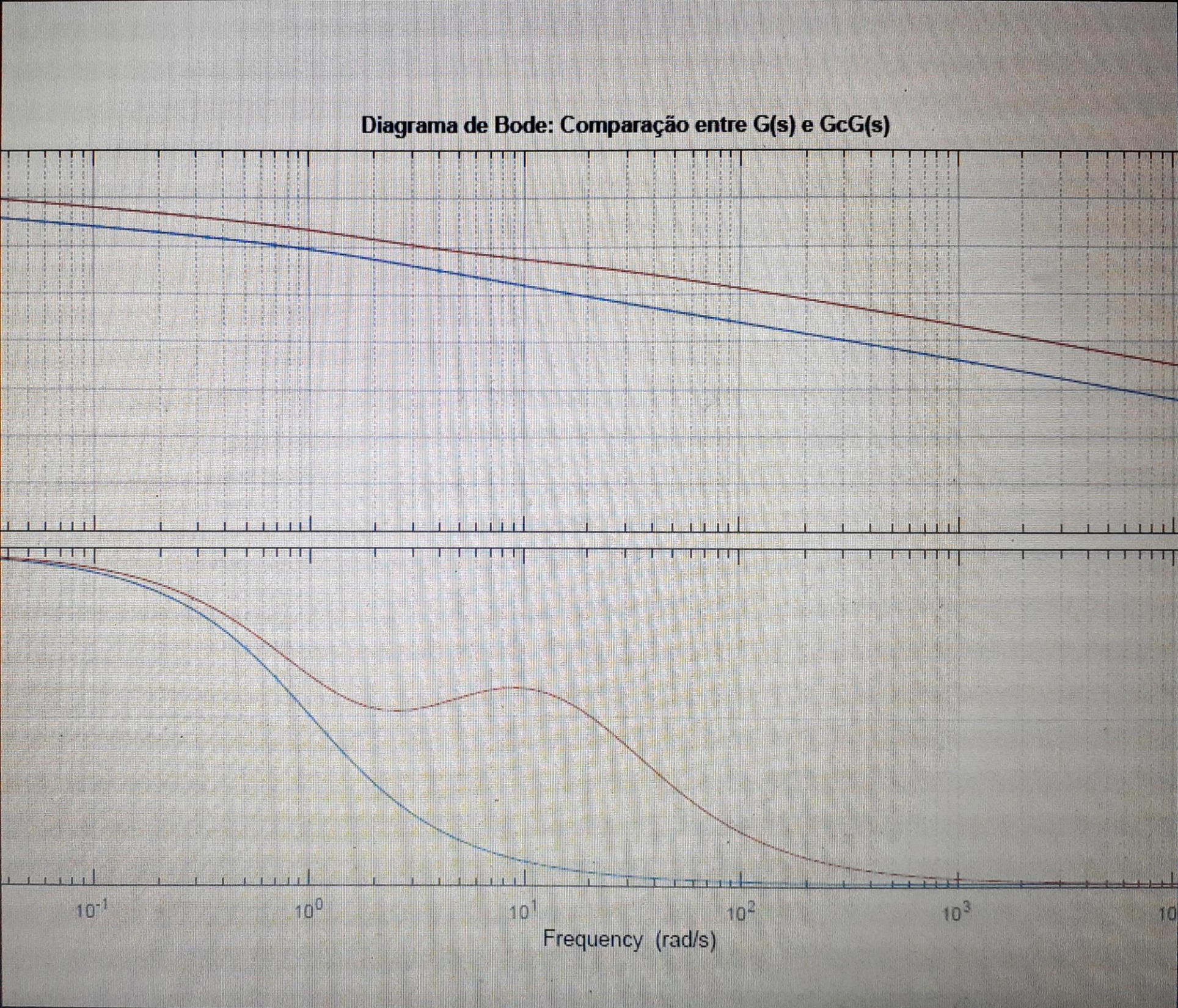
$$\text{iv)} \begin{cases} 20\log_{10}\left|G_1(j\omega)\right| = 0\ dB \Rightarrow \dfrac{50}{\omega\sqrt{\omega^2+1}} = 1 \Rightarrow 50 = \omega\sqrt{\omega^2+1} \Rightarrow \omega_c = 7.036\ rad/s \\ 0 - 90° - \operatorname{tg}^{-1}\left(\dfrac{\omega_c}{1}\right) = -171.91° \approx -172° \Rightarrow \varnothing = 180° - 172° = 8° \Rightarrow \varphi = 50° - 8° + 5° = 47° \\ \operatorname{sen}\varphi = \dfrac{1-\alpha}{1+\alpha} \Rightarrow \alpha = 0.155 \Rightarrow \left|G_1(j\omega_m)\right| = -20\log_{10}\left|\dfrac{1}{\sqrt{\alpha}}\right| = -8.092\ dB \end{cases}$$

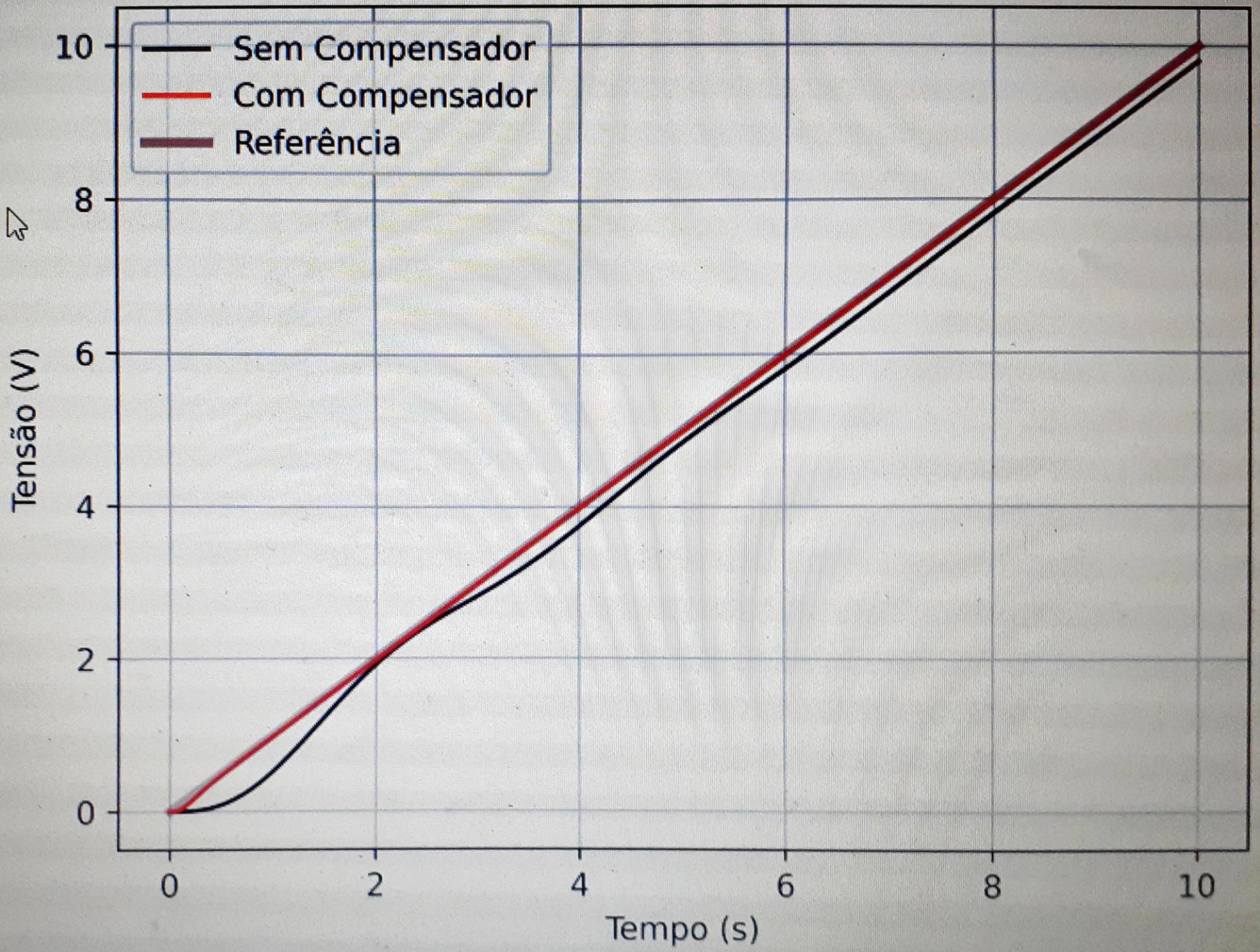
$$\text{v)} \begin{cases} 20\log_{10}\left|\dfrac{50}{\omega\sqrt{\omega^2+1}}\right| = \left|G_1(j\omega_m)\right| \Rightarrow \boxed{\omega_c = 11.244\ rad/s} \\ G_c(s) = K_c\dfrac{s+\frac{1}{T}}{s+\frac{1}{T\alpha}} \Rightarrow \boxed{G(s) = 64.447\dfrac{s+4.429}{s+28.545}} \end{cases}$$

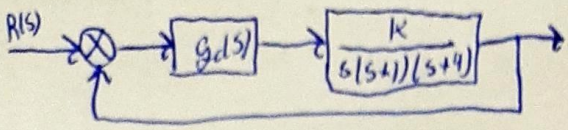
$$\begin{cases} \omega_1 = \dfrac{1}{T} = \omega_c\sqrt{\alpha} = 4.429\ rad/s \\ \omega_2 = \dfrac{1}{\alpha T} = \dfrac{\omega_c}{\sqrt{\alpha}} = 28.545\ rad/s \\ K_c = \dfrac{K}{\alpha} = 64.447 \end{cases}$$




Diagrama de Bode de G1(s)
Magnitude (dB)
100
50
0
-50
Phase (deg)
-90
-135
-180
10^-2
10^-1
10^0
10^1
10^2
Frequency (rad/s)


Diagrama de Bode: Comparação entre G(s) e GcG(s)
10-1
100
101
102
103
Frequency (rad/s)


Sem Compensador
Com Compensador
Referência
Tensão (V)
Tempo (s)
0
2
4
6
8
10

③ 

OBS.: Considerando $\gamma = \beta$

a) atraso-avanço, $\alpha = \beta$ e $K_c = 1$, $MF \geq 50°$, $Mg \geq 10dB$, $K_v = 10\,s^{-1}$, 5° ângulo do controlador

i $\left\{ G_c(s) = K_c \frac{\left(s+\frac{1}{T_1}\right)}{\left(s+\frac{\gamma}{T_1}\right)} \frac{\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{1}{\beta T_2}\right)} \right.$; $\gamma = \beta \Rightarrow G_c(s) = \frac{\left(s+\frac{1}{T_1}\right)}{\left(s+\frac{\beta}{T_1}\right)} \frac{\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{1}{\beta T_2}\right)} \cdot 1$

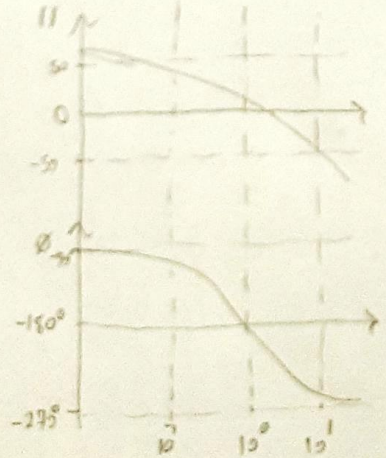
ii $\left\{ G_1(s) = \frac{40}{s(s+1)(s+4)} \Rightarrow G_1(j\omega) = \frac{40}{\omega\sqrt{\omega^2+1}\sqrt{\omega^2+16}} \angle 0° - 90° - tg^{-1}\left(\frac{\omega}{1}\right) - tg^{-1}\left(\frac{\omega}{4}\right) \right.$

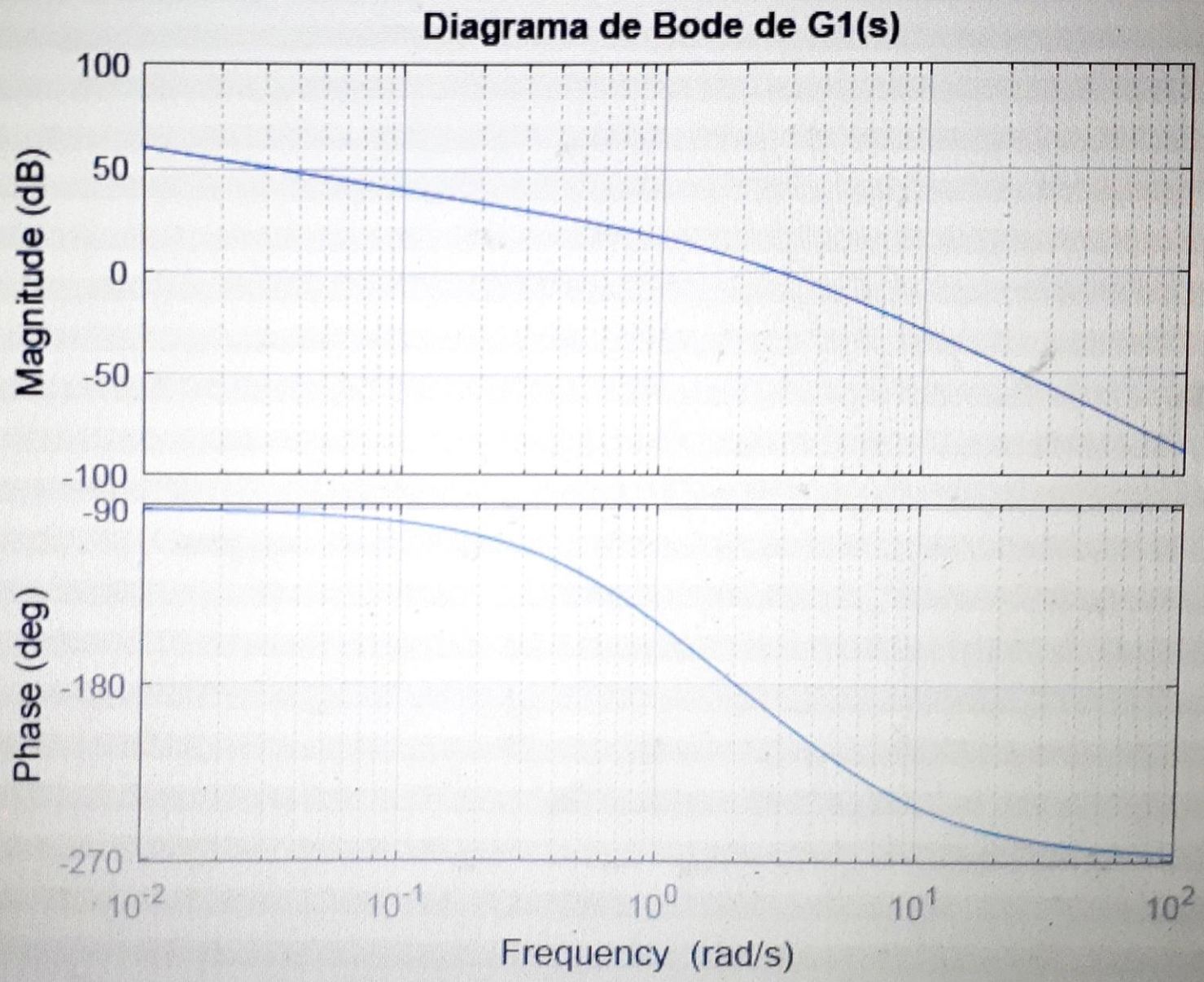
iii $\left\{ K_v = \lim_{s\to 0} \left[K_c \frac{\left(s+\frac{1}{T_1}\right)}{\left(s+\frac{\beta}{T_1}\right)} \frac{\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{1}{\beta T_2}\right)} \cdot \frac{K}{s(s+1)(s+4)} \right] \Rightarrow 10 = 1 \cdot \beta \cdot \frac{1}{\beta} \cdot \frac{K}{4} \Rightarrow \boxed{K = 40} \right.$

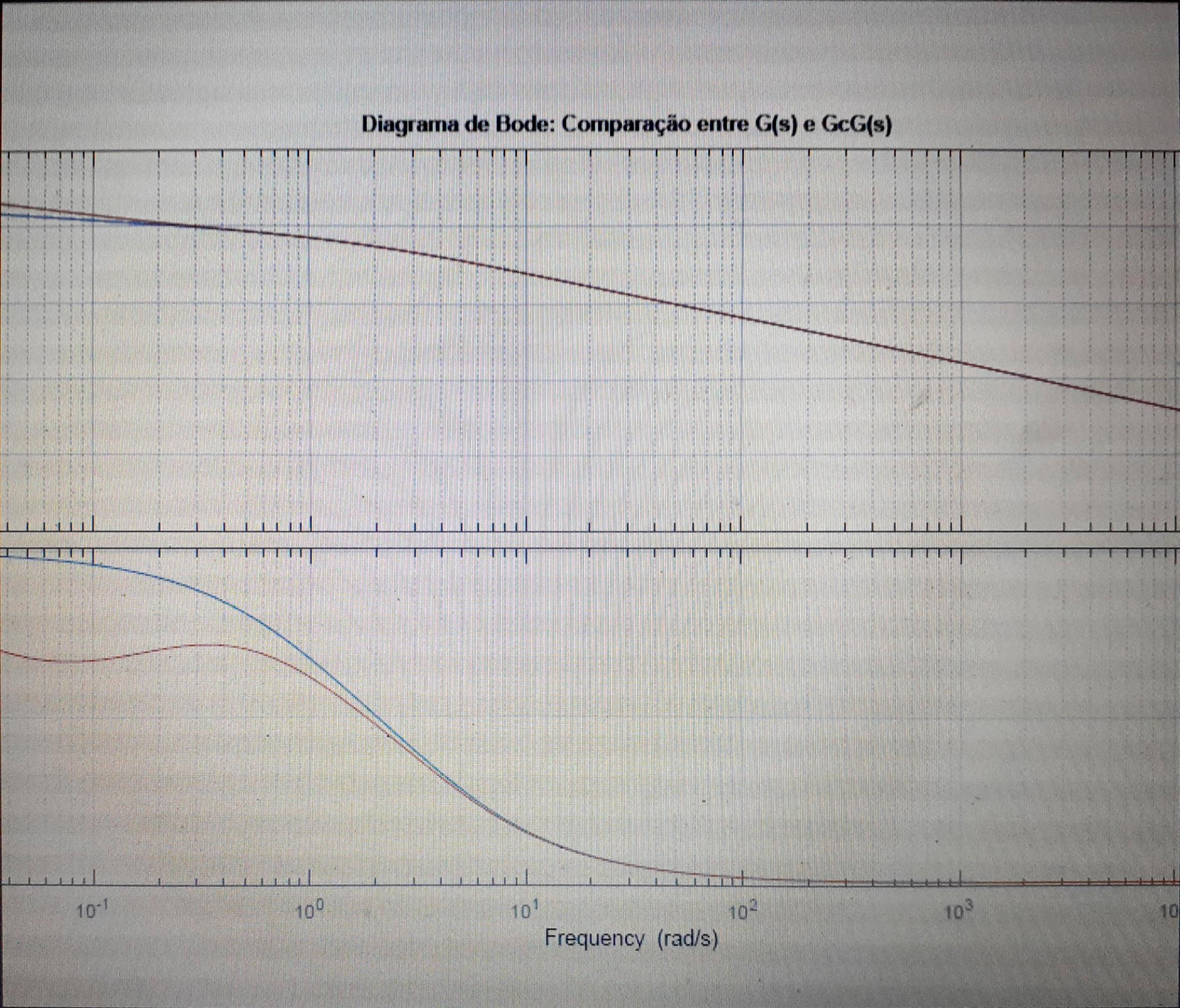
iv $\left\{ \begin{array}{l} \angle G_1(s) = -90° - tg^{-1}(\omega) - tg^{-1}\left(\frac{\omega}{4}\right) = -180° \Rightarrow \boxed{\omega_{mg} \cong 2 rad/s} \\ \varphi = 50° + 5° = 55° \Rightarrow \operatorname{sen}\varphi = \frac{\beta-1}{\beta+1} \Rightarrow \boxed{\beta = 10.059} \\ \boxed{\frac{1}{T_2} = \frac{\omega_{mg}}{10} = 0.200} \qquad \boxed{\frac{1}{T_2\beta} = 0.020} \end{array} \right.$

v $\left\{ \begin{array}{l} P = \left(\omega_{mg}, -20\log_{10}\left|G_1(j\omega_{mg})\right|\right) = \left(2\frac{rad}{s}, -6.021\,dB\right) \\ ax + b = y \Rightarrow b = -6.021 - 20\log_{10}(2) = -12.042 \Rightarrow y = 20\log_{10}(\omega) - 12.042 \quad [dB] \\ \left\{ \begin{array}{l} 0 = 20\log_{10}|\omega| - 12.042 \Rightarrow \boxed{\frac{\beta}{T_1} = 4} \\ -20\log_{10}|10| = 20\log_{10}|\omega| - 12.042 \Rightarrow \boxed{\frac{1}{T_1} = 0.400} \end{array} \right. \end{array} \right.$

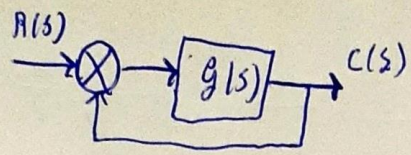
$$\boxed{G_c(s) = 1 \cdot \frac{(s+0.4)}{(s+4)} \cdot \frac{(s+0.2)}{(s+0.02)}} \quad //$$




Diagrama de Bode de G1(s)
Magnitude (dB)
100
50
0
-50
-100
Phase (deg)
-90
-180
-270
10^-2
10^-1
10^0
10^1
10^2
Frequency (rad/s)


Diagrama de Bode: Comparação entre G(s) e GcG(s)
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Frequency (rad/s)

④



a) $e_{ss} = ?$ para uma rampa unitária

Pelo gráfico, pode-se realizar as seguintes aproximações:

De 0.1 rad/s até 1 rad/s: decrescimento de -40 dB

logo pode-se concluir que a inclinação do reta é -20 dB por década.

Pelo gráfico podemos encontrar a seguinte ponto $P(0.1, 40)$

$b = 40 + 20 \log_{10}|0.1| = 20 \Rightarrow \boxed{y = -20 \log_{10}|w| + 20}$

Ao prolongar a reta, obtemos: $y = -20 \log_{10}|w| + 20 = 0 \Rightarrow \boxed{w = 10 \text{ rad/s}}$

$K_v = w_1 \Rightarrow \boxed{K_v = 10\ s^{-1}}$

$e_{ss} = \frac{1}{K_v}$

b) $e_{ss} = ?$ para degrau unitário

Observando o gráfico, quando $w \to 0$, o ganho tende a infinito, sendo assim esse sistema não é do tipo 0, logo: $e_{ss} = 0$.

Portanto, para $e_{ss} = 0$, K_v deve ser infinito.

$e_{ss} = \frac{1}{1 + K_v}$